Master in Physics of the Universe

Stellar Astrophysics: MESA Tutorial

Modules for Experiments in Stellar Astrophysics

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12/12/2025

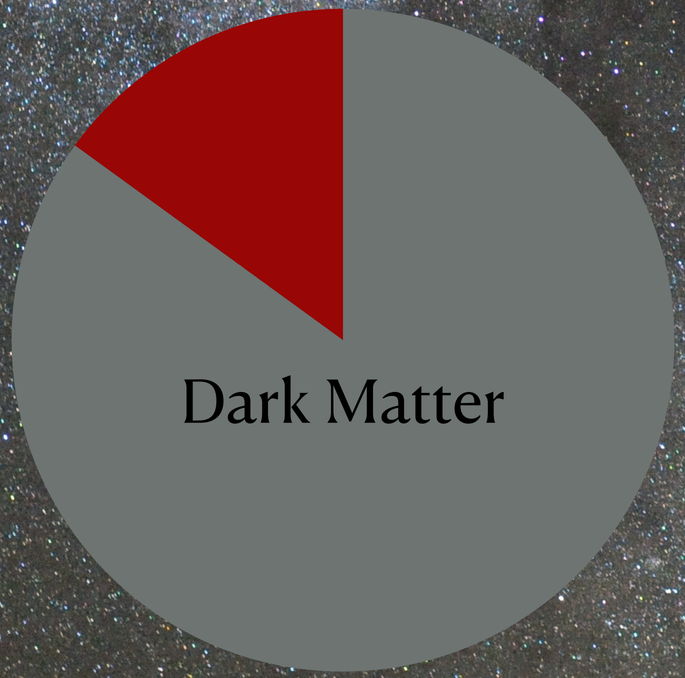
francandon@unizar.es

CAPA

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Universidad Zaragoza

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Axions??

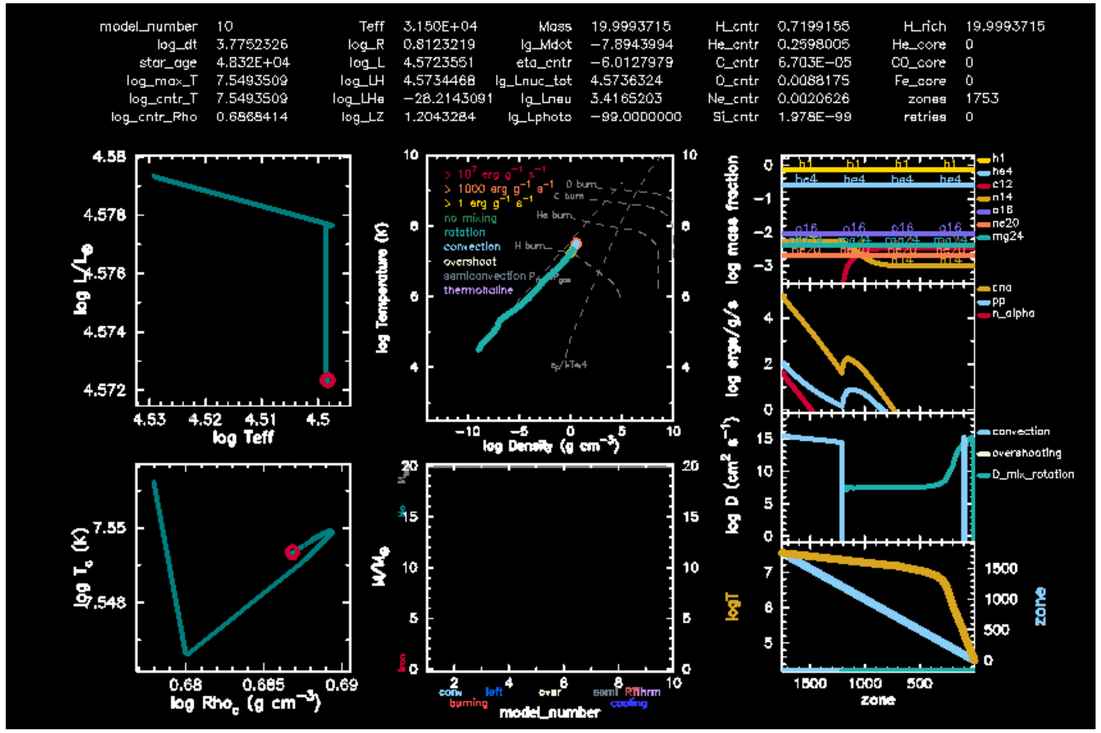
What makes you valuable?



0. Introduction to MESA

MESA

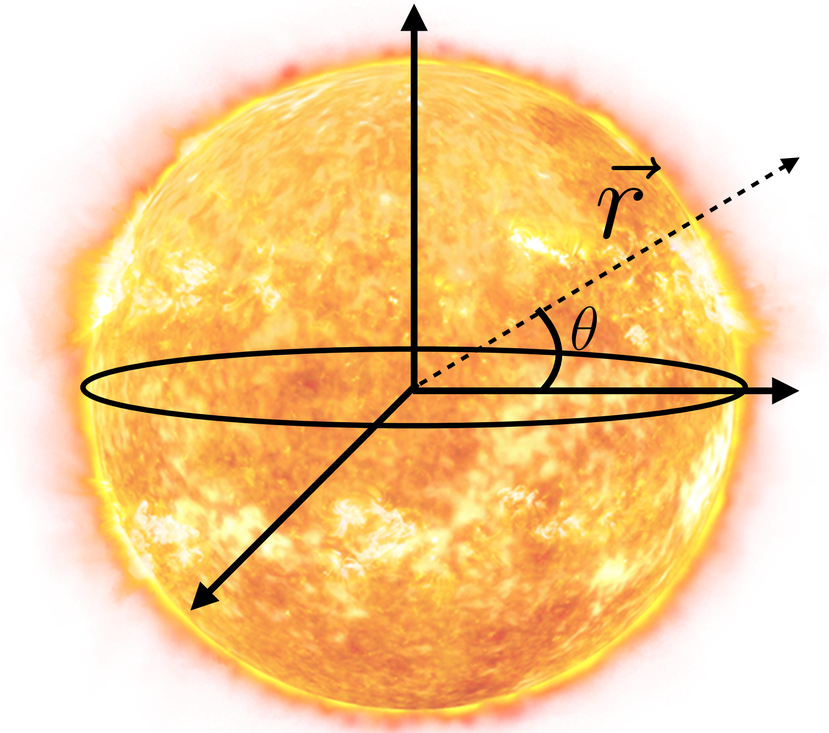
Modules for Experiments in Stellar Astrophysics

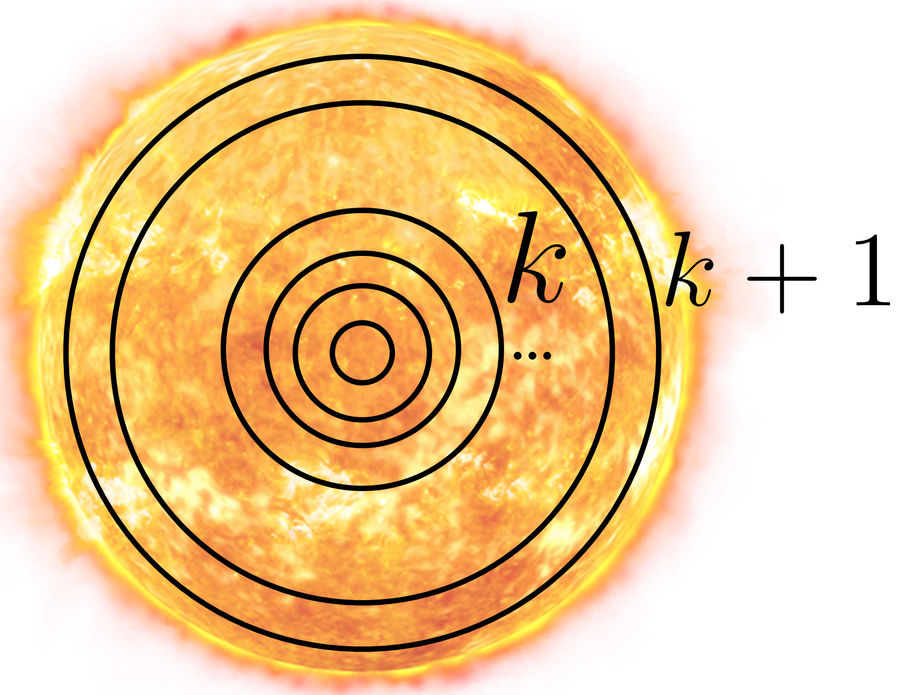


MESA is an advanced,
open-source 1D **stellar evolution code**

MESA Documentation

How does it work?





Each cell carries mass-averaged thermodynamic/composition variables

How does it work?

Hydrostatic Equilibrium

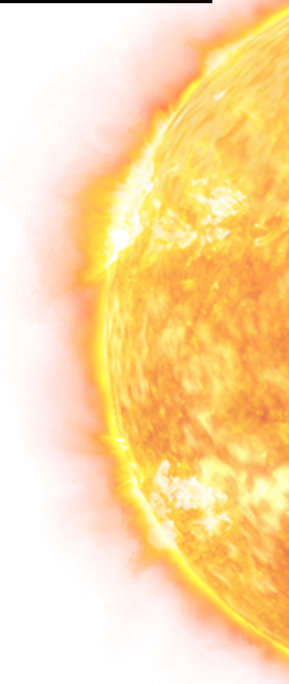
$$\frac{dP}{dr} = -\frac{G_N M_r \rho}{r^2}$$

Energy Conservation

$$\frac{dL_r}{dr} = 4\pi r^2 \epsilon \rho$$

Energy Transfer

$$L_r = \frac{4\pi r^2}{3\kappa\rho} \frac{d\left(aT^4\right)}{dr}$$



- Numerically solves the “standard” stellar evolution equations
- **Fast and efficient:** You can run it on your laptop and get publishable results!
- Very **flexible** to adapt to your specific problems
- **Detailed documentation** and direct interaction with the authors and the community via mailing list (daily updates!)

How does it work?

$$\frac{dm}{dr} = 4\pi r^2 \rho, \quad \frac{d}{dr} = \left(4\pi r^2 \rho\right) \frac{d}{dm}$$

Mass Conservation

$$\log r_k = \frac{1}{3} \log \left[r_{k+1}^3 + \frac{3}{4\pi} \frac{dm_k}{\rho_k} \right]$$

Momentum Conservation

$$P_{k-1} - P_k = dm_k \left[-\frac{Gm_k}{4\pi r_k^4} - \frac{a_k}{4\pi r_k^2} \right]$$

Energy Transport

$$T_{k-1} - T_k = dm_k \left[\nabla_{T,k} \left(\frac{dP}{dm} \right)_{\text{hydro}} \frac{T_k}{P_k} \right]$$

$\nabla_T = d\ln T / d\ln P$

Energy Conservation

$$L_k - L_{k+1} = dm_k \left(\epsilon_{\text{nuc}} - \epsilon_{\nu,\text{thermal}} + \epsilon_{\text{grav}} \right)$$

Composition Evolution

$$X_{i,k}^{n+1} - X_{i,k}^{n} = \underbrace{\left(\frac{dX_{i,k}}{dt} \right)_{\text{burn}}^{n+1} \Delta t}_{\text{nuclear network}} + \underbrace{\frac{\left(F_{i,k+1} - F_{i,k}\right)^{n+1} \Delta t}{\Delta m_k}}_{\text{mixing as diffusion}}$$

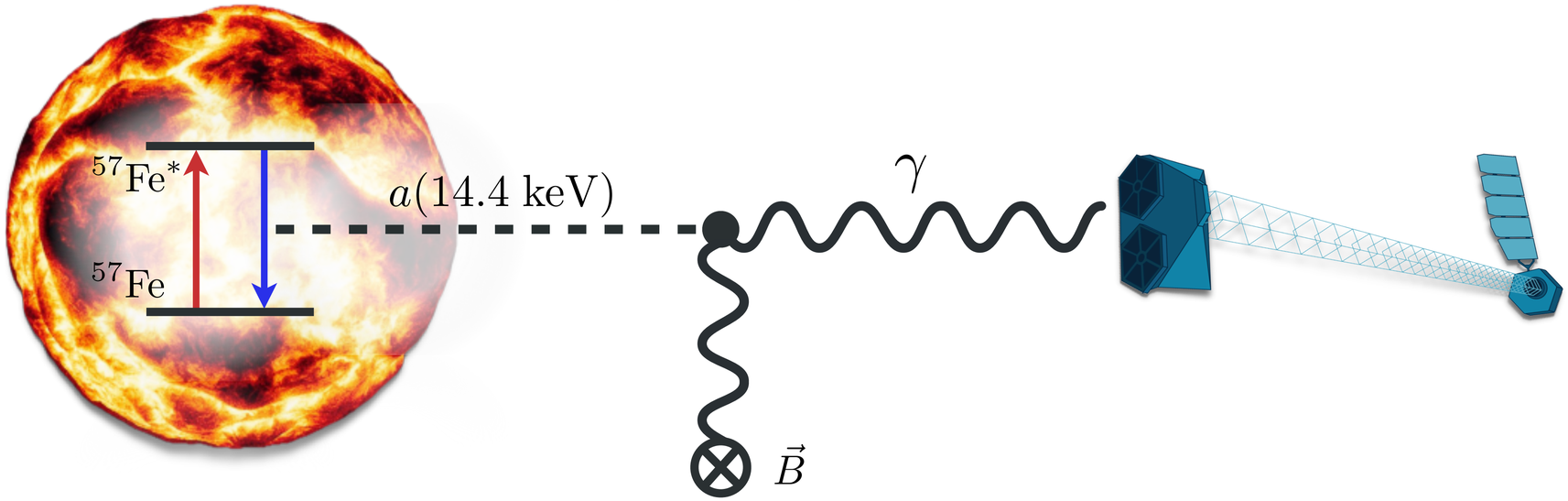
A home-made example

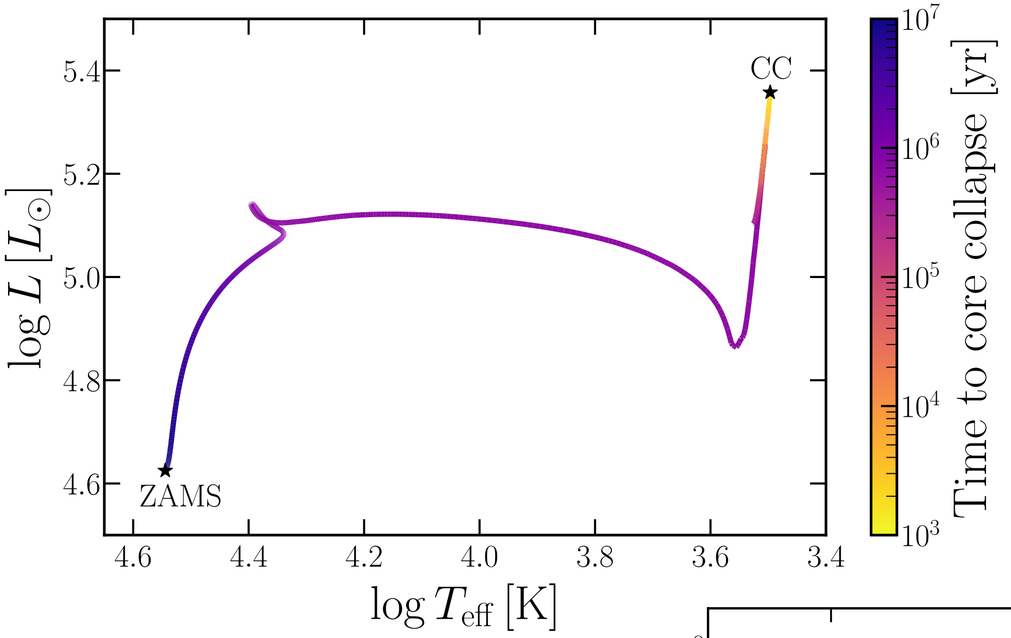
Probing the Axion-Nucleon Coupling with Supergiant Stars

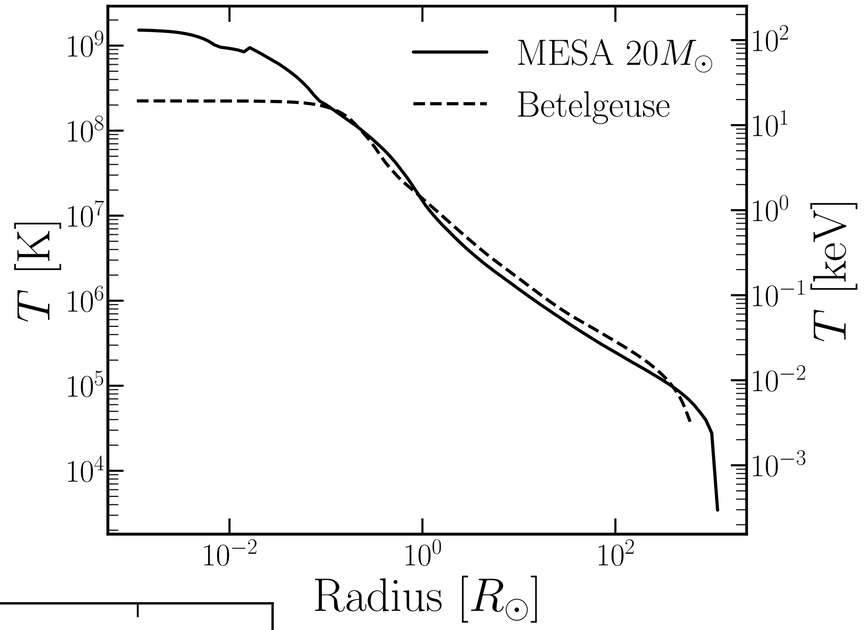
Francisco R. Candón [1,2] Pablo Casaseca [2] Maurizio Giannotti [2]
Mathieu Kaltschmidt [2] Jaime Ruz [1,2] and Julia K. Vogel [1,2]

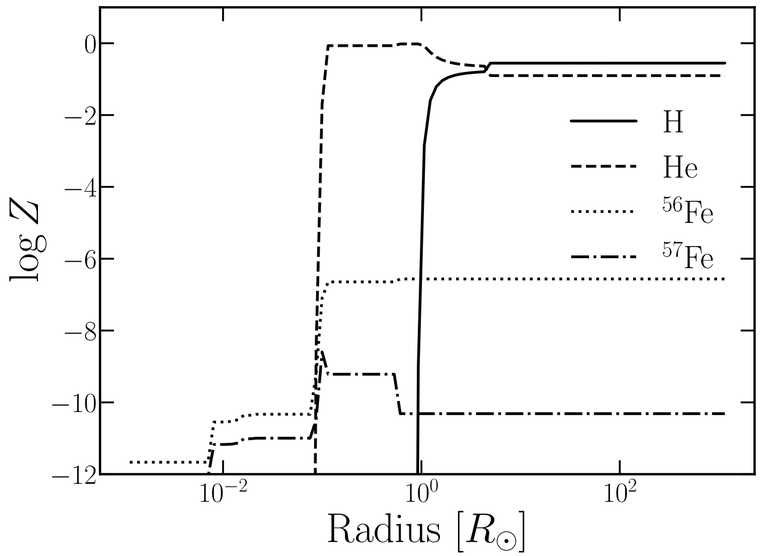
[1] *Fakultät für Physik, TU Dortmund, Otto-Hahn-Str. 4, Dortmund D-44221, Germany*
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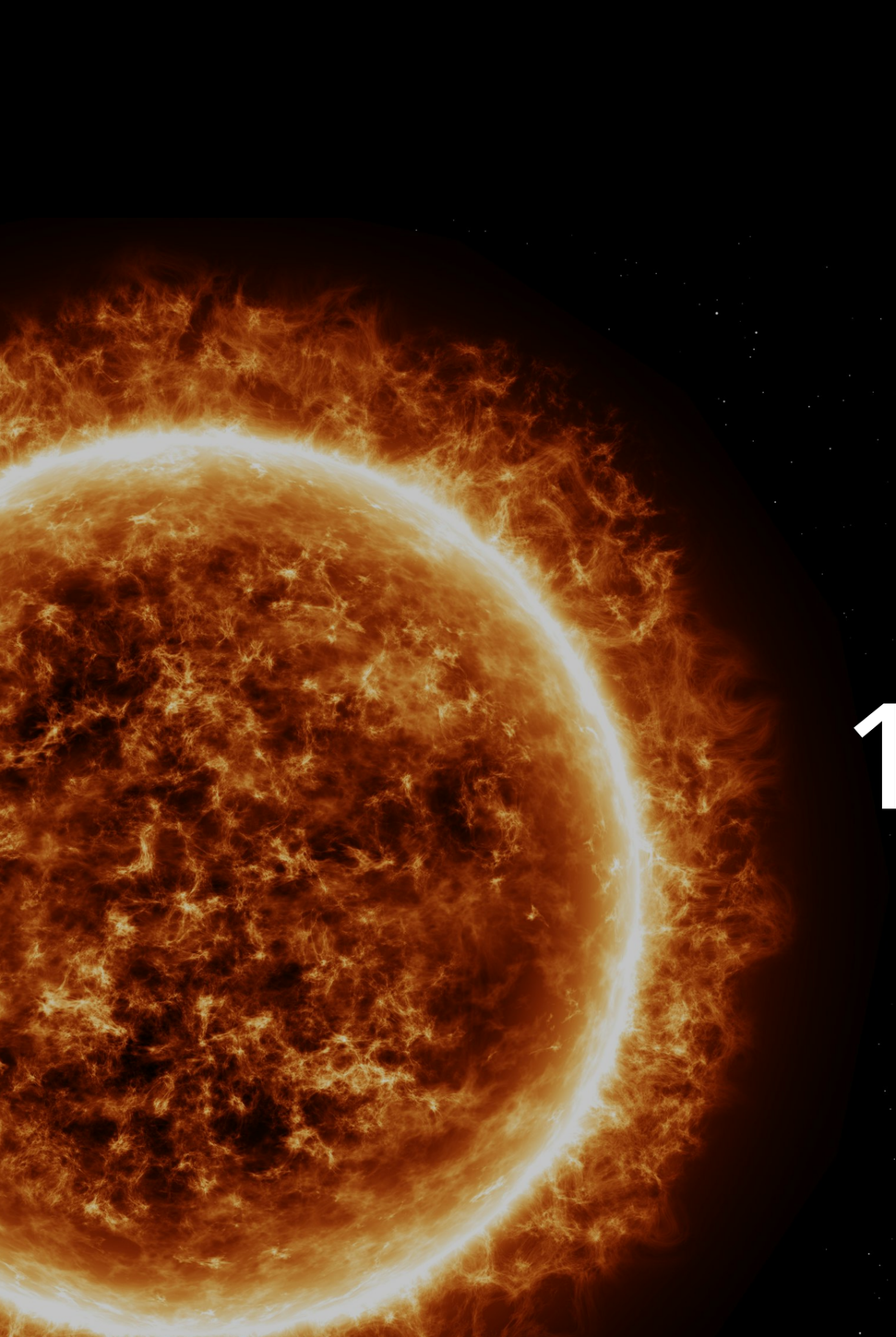
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CC
ZAMS
5.4
5.2
5.0
4.8
4.6
4.6
4.4
4.2
4.0
3.8
3.6
3.4
log L [$L_\odot$]
log $T_{\rm eff}$ [K]
10^7
10^6
10^5
10^4
10^3
Time to core collapse [yr]


MESA $20M_\odot$
Betelgeuse
10^9
10^8
10^7
10^6
10^5
10^4
T [K]
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
T [keV]
10^{-2}
10^0
10^2
Radius [$R_\odot$]


H
He
^{56}Fe
^{57}Fe
0
−2
−4
−6
−8
−10
−12
log Z
10^{-2}
10^0
10^2
Radius [$R_\odot$]



1.Getting Started

Getting started

We will start by running the minimal example provided by MESA

It can be found in

```
$MESA_DIR/star/work
```

and for today (and in general) I recommend to always copy it as your starting point:

```
cp -r $MESA_DIR/star/work path/to/your/cwd/name
```

The general workflow is then always the same, set up the simulation parameters and then run the scripts ./clean (remove any remnants from previous compilations), ./mk (compile the current setup) and finally ./rn (run the simulation)

Before running a simulation, we need to specify the details of our model of interest

The relevant changes can be made in the different input lists (“inlists”)

1. inlist_project

   Set up physics and numerics for your simulations

2. inlist_pgstar

   Set up the parameters for plotting

You can have multiple of these inlist files but you should always check that the main inlist in your cwd points to the right files!

Inlist

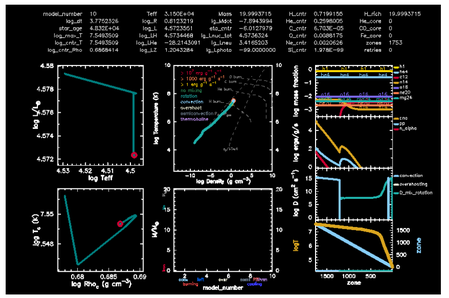
Inlist Project

Inlist Pgstar

Set up physics and numerics for your simulations

Metalicity, initial mass, nuclear network…

Set up the parameters for plotting



The output of your simulation will be stored in the subfolder LOGS/ in your cwd

We distinguish between two main files that will be saved

1. LOGS/history.data

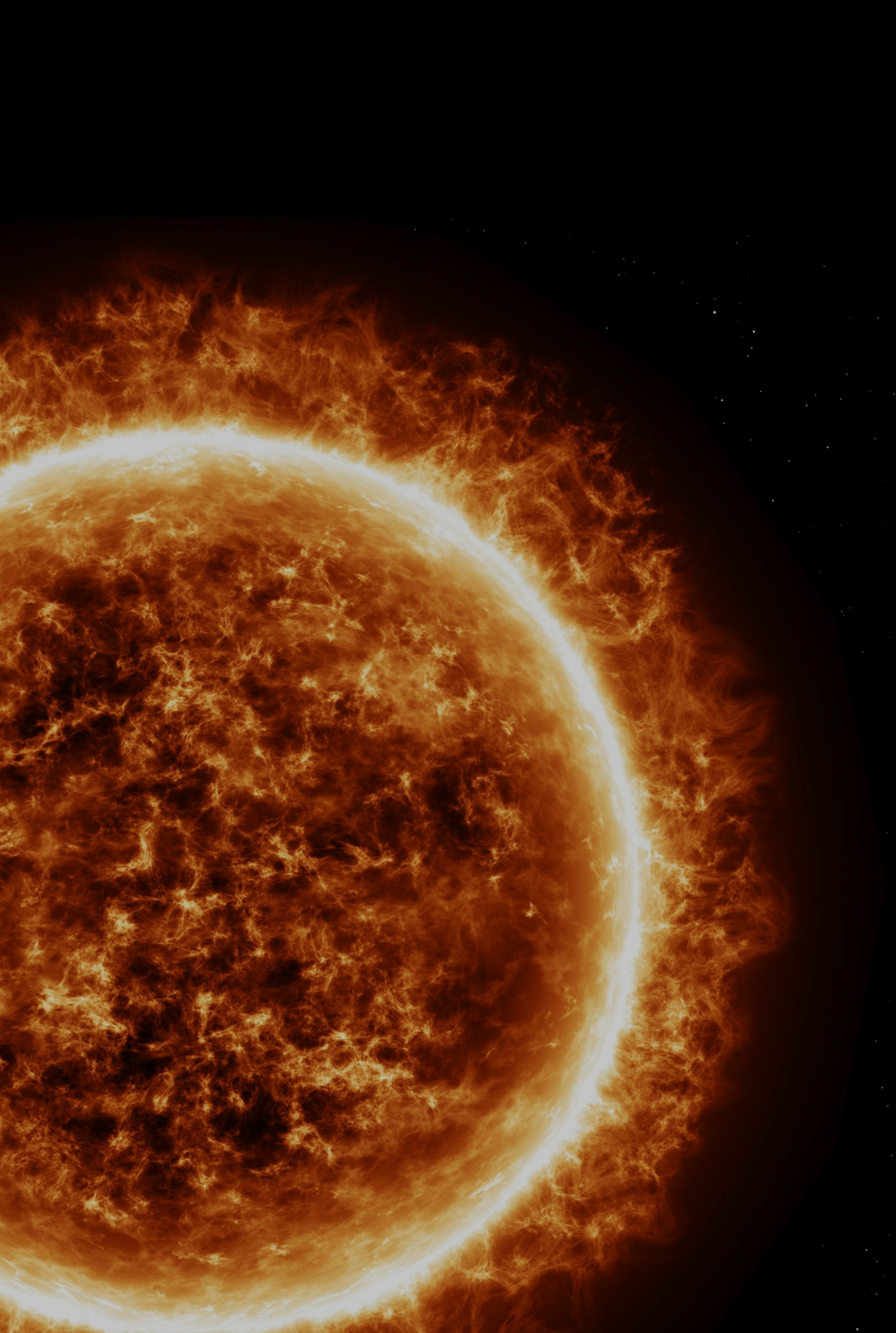
   One line per model with information such as age, luminosity, mass etc. so you can see how things change over time

2. LOGS/profileX.data

   One file per model X with the internal structure of each model, e. g. chemical composition profiles, temperature profile etc.

There are additional files, such as profiles.index or photos/ that can be useful to not loose track of the profiles for bigger simulations and for restarting simulations from a checkpoint

2. Simulating the Sun



- Diameter ~ 1.4 Million km
- Mass ~ 2 x 10^{30} kg
- Core Temperature ~ 15 Million °C
- Surface Temperature ~ 5600 °C
- Mostly made of H and He
- Main Sequence Star with Spectral Type G2V
- Age ~ 4.5 billion years

Setting up the Sun in MESA

How do we go from the setup in the minimal example to the Sun?

Hint: We need to look at three important characteristics! Any suggestions? :)

```
! starting specifications
  initial_mass = 1 ! in Msun units
  initial_z = 0.02
```

```
! when to stop
  max_age = 4.5d9
```

Additionally we make some changes to the numerical setup to speed up the simulation

```
! solver
   ! options for energy conservation (see MESA V, Section 3)
   energy_eqn_option = 'eps_grav'
   use_gold_tolerances = .false.
```

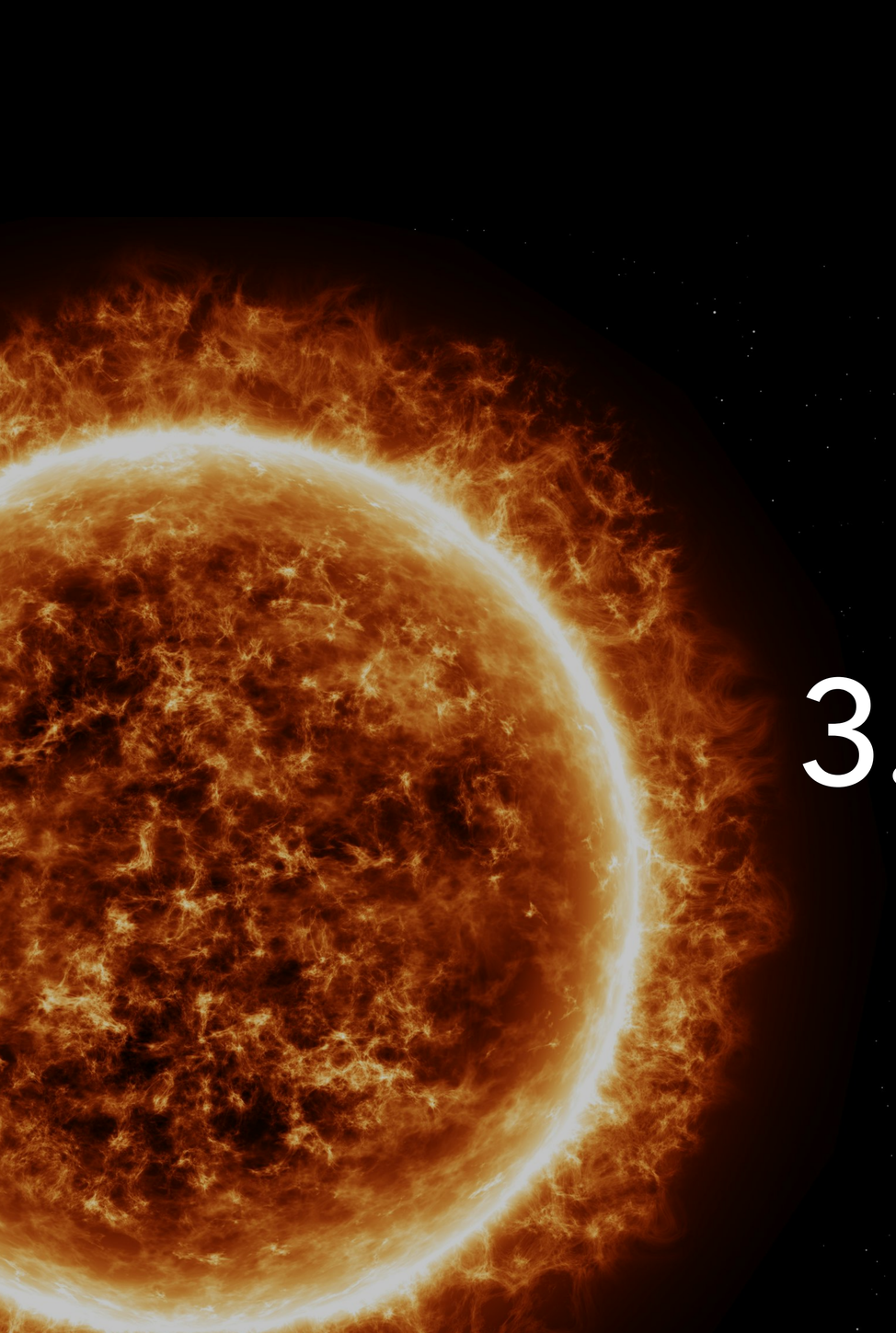
What could be problematic about this? Ref: MESA Publication

Controlling the PGStar Visual Output

- For this simulation use the sun_inlist_pgstar (make sure to point to this file in inlist!)
- Compared to the default values from $MESA_DIR/star/defaults/pgstar.defaults check that we use Grid1_file_flag = .true. to save the output as png files in the folder png/ (make sure it is empty by typing rm -r png/*)
- To produce a movie use

```
images_to_movie 'png/grid*.png' movie.mp4
```

- Look at the outputs of the simulation and the movie. Does the star you just simulated have the structure you expected for the Sun?



3. Analysis with python

Analysis with python

We are going to use the python package `py_mesa_reader` to load our output.

```
pip install mesa_reader
```

You can find the documentation and the GitHub repository

I prepared a Jupyter Notebook for you that you can find in our repository

`MESA_Tutorial1.ipynb`

Test other predefined models

You have access to a large amount of predefined models. See the list.

In order to work with them you have to create a copy in your working directory of the corresponding model folder located in

```
cd $MESA/star/test_suite/
```

For example, you can run a White Dwarf model just by copying the folder

```
cp -r $MESA/star/test_suite/make_co_wd path/to/your/
```

Look at the description in README.rst and understand the key parameters

Follow the usual routine to run the simulation (./clean then ./mk then ./rn)